

Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #10: Student Club Event Ticketing & Budget Portal

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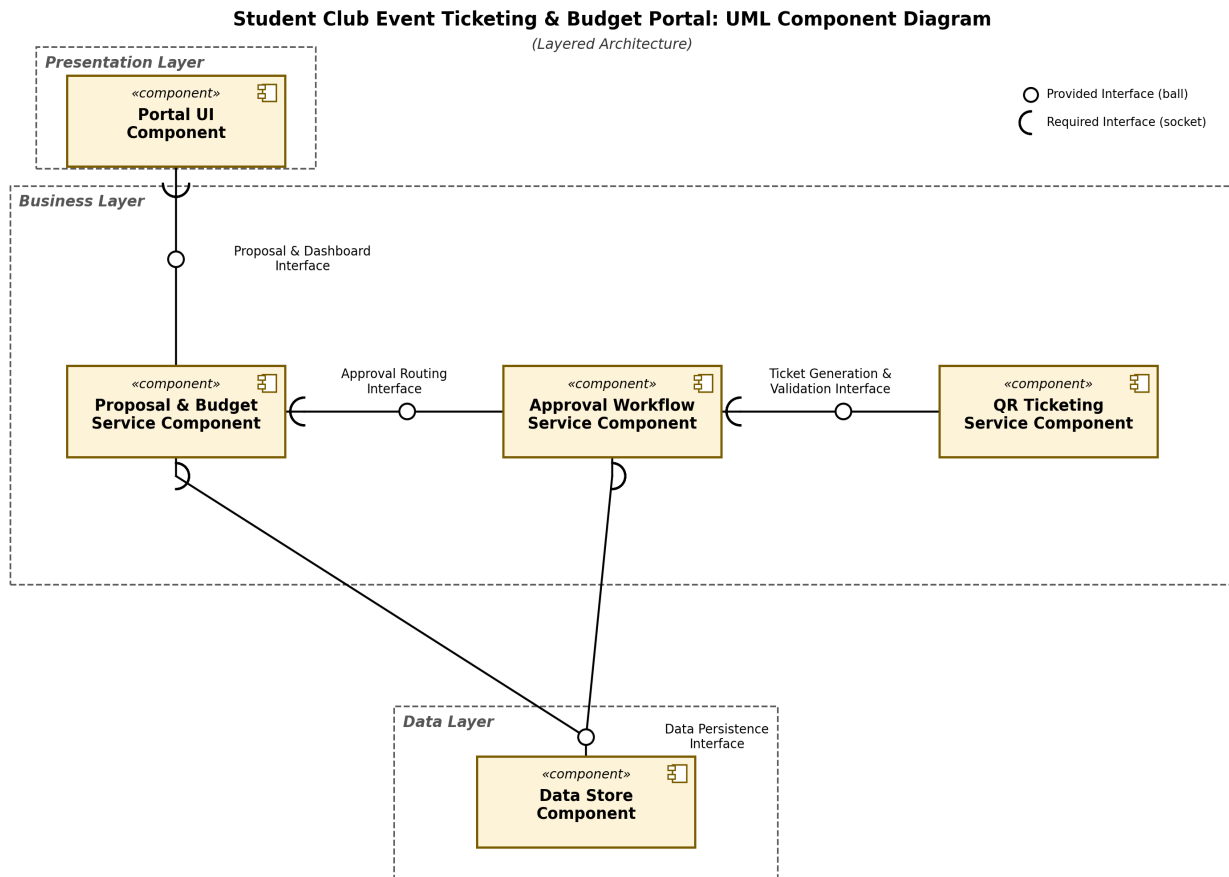
Class: SE Section B

Faculty: Dr. Sapna V M

Project: Student Club Event Ticketing & Budget Portal

1. UML Component Diagram

The diagram below shows the 5 core components of the Student Club Event Ticketing & Budget Portal organised into a Presentation Layer, a Business Layer, and a Data Layer, and the 4 required interfaces between them. Provided interfaces are shown with ball notation; required interfaces are shown with socket notation.



#	Component	Layer	Responsibility
1	Portal UI Component	Presentation	Web/app interface used by the Club Lead, Attendee, Faculty Coordinator, Finance Office, Dean and Campus Admin to submit proposals, register for or view tickets, and view the real-time status dashboard (FR-002, FR-005)
2	Proposal & Budget Service Component	Business	Captures event proposals, validates budget fields, tracks proposal status, and serves the live status/budget feed for the dashboard (FR-002, FR-005)
3	Approval Workflow Service Component	Business	Routes the proposal sequentially through Faculty Coordinator, then Finance Office, then Dean sign-off (FR-001)
4	QR Ticketing Service Component	Business	Generates a unique QR-code ticket per approved attendee registration, and validates or rejects tickets at check-in, flagging duplicate scans (FR-003, FR-004, NFR-001, NFR-002)
5	Data Store Component	Data	Persists proposal, approval-history, budget and ticket data; the sole point of database access for the whole system

Requirement Traceability

Every functional and non-functional requirement from the Lab 1 requirements table is covered by at least one component above; no new component was needed beyond the 5 shown.

Requirement	Covered By
FR-001: Multi-stage budget approval routing	Approval Workflow Service Component
FR-002: Club Lead submits event proposal	Portal UI Component, Proposal & Budget Service Component
FR-003: Generate unique QR ticket on approval	QR Ticketing Service Component
FR-004: Campus Admin scans ticket, rejects duplicates	QR Ticketing Service Component
FR-005: Real-time status/budget dashboard	Portal UI Component, Proposal & Budget Service Component
NFR-001: Sub-100ms ticket validation	QR Ticketing Service Component
NFR-002: 500 concurrent ticket registrations	QR Ticketing Service Component

2. Architectural Justification

Architecture Selection: We chose the **Layered Architecture** for the Student Club Event Ticketing & Budget Portal.

Reason 1: Sequential Approval Maps Naturally onto Layers

FR-001 requires proposals to move through a strict sequence of sign-offs, Faculty Coordinator, then Finance Office, then Dean, before any funds are reserved. This sequential validation and hand-off pattern maps directly onto a layered model, where a request passes through well-defined stages in the Business Layer before it ever reaches the Data Layer. Keeping this workflow logic inside the Approval Workflow Service Component, with a single shared Data Store beneath it, keeps proposal status and reserved budget atomically consistent, something that is far harder to guarantee once approval state and budget state live in separately-owned databases.

Reason 2: Simplicity Matched to the Actual Scale

The portal serves a single campus and a small, well-defined set of actor types (Club Lead, Attendee, Faculty Coordinator, Finance Office, Dean, Campus Admin), with a moderate concurrency requirement of 500 simultaneous ticket registrations (NFR-002). This is not an internet-scale system, so it does not need independently deployed, independently scaled services. A Layered Architecture keeps development, testing and deployment simple for a single team, avoiding the operational overhead of running and coordinating multiple separate services for a workload a well-structured application can comfortably handle on its own.

Security Advantage

In this layered design, the Data Layer is never reached directly by the Presentation Layer or by external clients; every read or write to proposal, approval, budget and ticket records passes through the Business Layer's Data Persistence Interface. This gives the system one centralised point where authentication and authorisation checks are enforced consistently, which is a smaller, easier to reason about attack surface than an architecture where several independent services each maintain their own path to the data.

Performance Benefit

Because the QR Ticketing Service Component sits in the same Business Layer as the Approval Workflow Service Component rather than behind a network call to a separately deployed service, it can hold the current event's issued-ticket status in memory instead of hitting the database on every scan. In-process access is inherently faster than a network hop, so this lets QR-code ticket validation at the venue complete well within the 100 ms target (NFR-001) even during high-footfall check-in periods, since most validation requests are served from the business layer's cache rather than requiring a round trip to the Data Layer for each individual scan.